Sathwik P. Kallapur

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Full and unrestricted working rights in Australia (Subclass 485 visa holder)

PROFESSIONAL SUMMARY

Backend and systems software engineer with almost 3 years of commercial experience in distributed platform orchestration, high-throughput microservices, and performance engineering. Proven track record across scalable platforms: led a Go migration achieving a 50x speedup and cutting cloud costs at scale (Latlong), built a geocoding engine 2x more accurate than Google Maps for logistics routing, automated server and platform deployments to relieve broadcast engineers of complex infrastructure burdens (NEP Australia), and rectified financial ledger drift to guarantee transaction precision for finance teams (CoffeeBeans). Brings a language-agnostic mindset and low-level systems instincts (C++, concurrency, profiling), leveraging agentic workflows to maximise delivery velocity across tech stacks.

WORK EXPERIENCE

Orchestration Software Engineer April 2026 – Oct 2026
NEP Australia *Melbourne, Australia*

- Built distributed Go workers to automate Canonical MaaS server deployments via Conductor workflows, replacing ∼1.5 hours of manual provisioning per live event and freeing broadcast engineers from setup overhead.
- Engineered automated E2E test pipelines and deployment lifecycles using Robot Framework, Jenkins, and Python, ensuring early defect detection and high operational reliability for live production teams.
- Built custom Linux OS images with MaaS Packer and Ansible for specialised production broadcast environments.
- Conducted peer code reviews on release branches, enforcing architectural standards, Go idioms, and test coverage to safeguard live broadcast pipeline stability.
- Leveraged Agentic AI tools (Codex, Antigravity) to accelerate developer tooling, test generation, and architecture spikes, increasing feature velocity for platform engineering teams.

Software Development Engineer (SDE) 1 July 2023 – Dec 2023
Medpho (Telimedic Technologies) *Remote, India*

- Developed backend RESTful APIs and an AI-powered conversational service using Node.js and Dialogflow, enabling automated query resolution and appointment bookings for patients in regional towns.

Software Development Engineer (SDE) 1 July 2022 – July 2023
CoffeeBeans *Bengaluru, India*

- Refactored financial calculation logic with a custom precision type, eliminating rounding drift and ensuring 100% transaction ledger accuracy for finance teams and merchant partners.
- Implemented payment processing (Stripe, Razorpay) and optimised event-driven Kafka workers on AWS, improving throughput and checkout reliability for high-volume e-commerce shoppers.
- Reduced service latency by up to 30% and boosted throughput via Go runtime profiling (pprof) and concurrency tuning across AI recommendation and e-commerce services.
- Engineered CI/CD delivery pipelines (Jenkins, ArgoCD) and administrative internal tools, translating stakeholder requirements into clean APIs and accelerating deployment cadence for cross-functional engineering teams.

Software Engineering Intern Sep 2021 – June 2022
Latlong (ONZE Technologies) *Bengaluru, India*

- Led backend API migration from Ruby on Rails to Go, achieving ∼50x faster execution and ∼20x lower memory usage, cutting cloud costs and improving real-time API response for ride-hailing and food delivery clients.
- Designed and developed a Geocoding algorithm 2x more accurate than Google Maps API for local addresses, enhancing door-to-door routing and ETA precision for delivery drivers and logistics dispatchers.
- Developed core geospatial RESTful APIs in Go (Reverse Geocoding, Landmarks, Driving Directions with ETA) using PostGIS, serving real-time location queries under high request volumes.

SKILLS & CORE COMPETENCIES

Languages: Golang, C/C++, Python, Java, JavaScript, TypeScript, Ruby
Distributed Systems & Backend: RESTful APIs, Concurrency Patterns (Go), Orchestrator-Worker Pattern, Kafka Event-Driven Architecture
Databases & Geospatial: PostgreSQL, PostGIS, OracleDB, MongoDB, Redis
Cloud, DevOps & Infrastructure: Canonical MaaS, Linux (Ubuntu/Arch), Docker, AWS, OCI, Ansible, Packer, CI/CD (Jenkins, ArgoCD), Git
Testing & AI Tooling: Robot Framework, Allure, pprof, E2E/Mutation Testing, Antigravity, Cursor, Codex
SDLC & Business Competencies: Requirements Gathering, Feasibility Analysis, Systems Design, Agile Delivery, Code Reviews, Technical Documentation

PROJECTS

Racing Line Mapper | *Golang, Reinforcement Learning, OpenCV* 2025

- Reinforcement Learning system mapping optimal racing lines using MDP and image-processed track skeletons.
- Developed a custom arcade physics engine and transformed raw track maps into semantic Frenet frames.

SafeTrek | *FastAPI, Vue3, PostGIS, Docker* 2025

- Interactive road safety analysis platform using geographic visualisation to support urban policy decisions.
- Implemented complex geospatial queries in PostGIS and containerised the entire stack for deployment.

Generic Red-Black Trees | *C++ Template Programming* 2021

- STL-compatible RBT implementation with bidirectional iterators and custom predicate support.

MacOS Style Portfolio | *Vue3, Vite, JavaScript, Responsive UI* 2025

- Interactive web recreation of macOS GUI featuring window management and desktop app emulation.
- Engineered modular Vue3 components showcasing frontend design polish and responsive state management.

EDUCATION

Monash University Feb 2024 – Dec 2025
Master of Information Technology (MIT) *Clayton, Australia*

- GPA: 3.31/4.0. Key units: Cloud Computing, Advanced DBMS, Discrete Optimisation.
- US equivalent GPA: 3.88/4.0. (Calculated via WES iGPA tool)

PES University July 2018 – July 2022
Bachelor of Technology in Electronics and Communications *Bengaluru, India*

- Major GPA: 8.16/10. Minor in Computer Science (GPA: 8.25/10).
- Coursework: Data Structures, Image Processing, Computer Networks, Generic Programming, Operating Systems.

INTERESTS

High Performance Software Design, Systems Programming in Rust, Game Development, Physics Engines, Sim Racing, Science Fiction, Guitar, Rock Music, Martial Arts, Sports, and Mindfulness.